

A counterexample to the unrestricted bound in Conjecture 3464

Chengzhe Gao

The English and Chinese versions both assert the universal bound $\lambda(G) \leq \Delta(G)^2$, without a lower bound on Δ . Here λ is the least label range for an $L(2, 1)$ -labeling. This note disproves that unrestricted assertion. It does not disprove the classical version restricted to $\Delta \geq 2$.

Theorem 1. *For the single-edge graph K_2 , one has $\Delta(K_2) = 1$ and $\lambda(K_2) = 2 > 1 = \Delta(K_2)^2$.*

Proof. Each vertex has exactly one neighbor. In any valid labeling, the two adjacent vertices have labels differing by at least two. Hence their label range is at least two, regardless of how large the labels are. The labels zero and two attain range two. There are no vertices at distance exactly two, so the second distance condition is vacuous. $\square$

The later assertions in the source form part of a conjunction containing the displayed universal bound; a counterexample to this necessary part already disproves the conjecture as stated. No asymptotic claim is used. The omitted degree restriction is substantial: the standard formulation explicitly assumes $\Delta \geq 2$; see the statement in Discrete Applied Mathematics, DOI 10.1016/j.dam.2025.10.011.

Formal verification. The Lean file defines finite simple graphs with symmetry and no loops, computes degrees from the entire vertex list, and defines distance two as nonadjacency together with a common neighbor. The definition `L21` imposes both standard separation conditions. `SpanAtMost f R` means that every label lies in some translated interval $[\ell, \ell + R]$. The theorem `span_range_equivalence` proves that this agrees with maximum minus minimum on this graph. `every_valid_span_at_least_two` quantifies over all natural labels and all intervals; it is not a search with an imposed label bound. The labels zero and two and this universal lower bound yield `edge_minimum_span`. Finally, `conjecture3464_false` negates the claimed universal bound itself. Integer labels give the same minimum range by translating a finite labeling so that its minimum is zero.

The program uses Lean 4.19.0 and `Std`. The printed axiom lists contain only Lean's standard logical foundations; no unproved declaration, external oracle, or native evaluation tactic occurs.